

ZAIRE WILSON

Software Engineer • Real-Time Simulation & XR • Full-Stack Applications

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PROFILE

Software engineer with 5+ years shipping production web applications, backend services, and real-time 3D simulation systems. Builds the full path from physical input to virtual state: motion-captured props driving a mixed reality environment, VR controller physics driving live gameplay, and custom networking synchronizing 200+ devices in a DOD mixed reality training system. Deep C#/.NET and Unity/Unreal background, with Python, TypeScript/React, Docker, and CI/CD across cloud and air-gapped deployments. Actively builds with LLM tooling (MCP integrations, context engineering) in personal work, treating AI as an iterative development tool grounded in solid engineering fundamentals.

TECHNICAL SKILLS

Simulation & 3D: Unity, Unreal Engine 5, Godot, proprietary engines, physics & collision systems, OpenXR/VR, mixed reality, Vicon motion capture, NavMesh pathfinding, procedural generation, Maya, Blender

Languages: C#, Python, TypeScript/JavaScript, C++, SQL, Java

Applications & Services: ASP.NET Core Web API, REST APIs, webhooks, React, WinForms, IIS, nginx, real-time networking, server-authoritative architecture

DevOps & Infrastructure: Docker, GitHub Actions, GitLab CI/CD, Jenkins, Ansible, Nexus, PowerShell, Git, DigitalOcean, Edgegap, air-gapped environments

Data: Microsoft SQL Server, Oracle, stored procedures, event telemetry and analytics instrumentation (Amplitude)

AI Tooling: LLM-assisted development, Model Context Protocol (MCP) integrations, prompt and context-window optimization (personal projects)

EXPERIENCE

Software Developer II | Scientific Research Corporation (DOD Contractor), Charleston, SC Dec 2024 – Present

- Implemented custom networking for real-time communication across 200+ dynamic devices in a mixed reality training environment, keeping physical hardware and virtual simulation state in sync.
- Built .NET (C#) web applications, WinForms desktop tools, and ASP.NET Web API services to NIST security standards and DOD compliance requirements.
- Built and maintained GitLab CI/CD pipelines for automated testing in an air-gapped classified facility, delivering reliably under restricted-network constraints.
- Designed SQL Server stored procedures, functions, and developer tooling that automated routine database operations; drove build and test automation across multiple technology stacks.

Lead Software Engineer (Contract) | Five Mics Studios, Remote Sep 2024 – May 2026

- Led engineering on a 4-person agile team, owning client, server-authoritative multiplayer, backend, and tooling end to end; shipped the demo behind a \$17K, 160+ backer crowdfunding campaign.
- Built an ASP.NET Core API gateway that provisions dedicated game servers on demand, using webhook-driven readiness checks and remote versioned configuration; deployed with Docker, GitHub Actions, and nginx.
- Designed a data-driven, event-based ability system (triggers → filters → actions) and a custom authoring tool that let non-engineers ship 50+ content items without code.

Gameplay Engineer | GOLF+ (VR Golf Simulation), Remote Feb 2022 – Sep 2024

- Built physics-based systems for a top-selling live-service VR golf simulation, including accidental-hit detection from tracked controller swing data and rules-accurate ball relief placement.
- Developed and integrated REST APIs for live multiplayer and real-time data synchronization across a 1,000+ daily user platform; built React interfaces on live backend services.
- Instrumented gameplay telemetry via the Amplitude API, built in-engine visualization tools for QA and product teams, and cut frame hitches and load times through multithreading and GC optimization.

Unity Developer | Meta Gaming Studios (now Makrö Gaming Studios), Kennesaw, GA Aug 2021 – Feb 2022

- Developed player-facing front-end UI and client-side gameplay features in Unity (C#) for digital slot titles.

SELECTED PROJECTS

Pandora's Box — Mixed Reality Escape Room | RPI CAVE Lab | Technical Excellence Award, RPI GameFest 2020 2020

- Co-developed a multiplayer mixed reality escape room in an immersive CAVE, using Vicon optical motion capture to track physical props and translate real-world manipulation into live virtual puzzle state.
- Integrated tracked physical objects with the simulated environment so players solved puzzles by hand rather than with controllers, a physical-to-digital loop analogous to sensor-driven digital twin synchronization.

Grocery Run VR | Unreal Engine 5, OpenXR, C++/Blueprints 2023 – Present

- Building autonomous NPC agents on a modular, data-driven state machine (idle, patrol, chase) with NavMesh pathfinding, event-driven perception of player actions, and runtime procedural level generation.

EDUCATION, CERTIFICATIONS & CLEARANCE

M.S., Critical Game Design | Rensselaer Polytechnic Institute, Troy, NY May 2021

B.S., Computer Science | Rensselaer Polytechnic Institute, Troy, NY | Undergraduate Researcher, CogWorks Lab May 2020

CompTIA Security+ (Active) **Active Secret Clearance** (T3, adjudicated 2024)